JENSEN PRODUCTION

Fleet Management System

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Phase 2 — Build Workbench

How the system will let you turn templates into bikes, end to end.

Prepared for: Dennis E. Jensen
Phase: 2 of 5
Status: Phase 1 (Parts catalog) is built and running.
May 2026

How to use this document

This is a follow-up to the Design Review you saw earlier. Phase 1 — the Parts catalog — is now built and running on the new system, and ready to replace your Excel. This document covers Phase 2: the Build Workbench, where templates become bikes.

It asks for your input on a small number of decisions where your answer changes how the system feels to use. Wherever you see a yellow box marked QUESTIONS FOR DENNIS, that's where we need your input.

Total time to read and respond: roughly 20 minutes. Mark the PDF up in any reader, scribble on a printout, or just answer in an email or call.

The big picture

The Build Workbench is what your mechanics will use to actually build bikes. It sits between the Parts catalog (Phase 1, done) and the Customer/Sales side (Phase 3, after this).

Three concepts.

Bike Template — a saved recipe. "To build a Hospital Service Bike, take these 47 parts in these quantities." You define each template once and reuse it for every order of that bike.

Manufacturing Order — a build directive. "Build 50 Hospital Service Bikes against template X, finish by August 1." An MO can be linked to a customer's sales order (in Phase 3) or stand alone for stock builds.

Bike — a single physical bike, with its own frame number and other identifiers. Each MO produces one or more bikes.

What the mechanic actually sees.

One Manufacturing Order at a time. They open it, check the parts list and stock, see the list of bikes (1 of 50, 2 of 50…), register the identifiers as each bike rolls off the line, and mark it complete. The system handles the bookkeeping — parts get drawn from stock automatically, the bike's status flips from "building" to "in stock," and when the last bike is done, the MO closes itself.

A walkthrough: "Aarhus Universitetshospital orders 50 service bikes"

Same scenario from the original Design Review — this time focused on what Phase 2 makes possible. Read it and tell us where it diverges from how things actually happen at Jensen Production.

Step 1 — The order arrives.

You receive an offer for 50 Hospital Service Bikes. The customer side (offers, sales orders, invoicing) lands in Phase 3. For Phase 2 we focus on the build itself, which is the part that actually changes how the workshop runs.

Step 2 — Open the workbench, create a Manufacturing Order.

You — or your lead mechanic — pick the template ("Hospital Service Bike, 53cm, white"), set the target quantity (50), planned start and finish dates, and the lead mechanic. You can either link the MO to a customer (Phase 3) or mark it as build-for-stock.

Step 3 — Stock check.

The template's parts list loads. Each row shows: part name, quantity per bike, total needed for this MO (qty per bike × 50), current on-hand. If anything is short, you see it before the build starts — not when a mechanic reaches for a part that isn't there.

Step 4 — Adjustments for this specific order.

The customer asked for an upgraded display unit. You substitute Display A for Display B on this MO. The system records the substitution. Past MOs against the same template still show the original recipe; a future MO against the same template will start from the original again unless you change the template itself.

Step 5 — Mechanics build, one bike at a time.

For each completed bike: click "Add bike" → fill in the identifiers required for this bike type. For an e-bike that's all five from your whiteboard: frame, lock, battery lock, battery, charger. The system validates each one (right format, not already used). Bike status: planning → building → in stock. Inventory gets drawn down automatically. The MO counter ticks: 1 of 50, 2 of 50…

Step 6 — Done.

When the 50th bike is registered, the MO closes itself. The bikes sit at status "in stock" until they're delivered (Phase 3). You can see the full build history per bike at any time.

Questions for Dennis

About 16 questions, organized by topic. They're roughly ordered by how much each one changes the design.

Templates and recipes

The recipe is the heart of the workbench. Get this right and the rest follows.

QUESTIONS FOR DENNIS — Templates

Q1. Roughly how many distinct bike templates do you have today — or expect to have? (5? 20? 50?)

Q2. When you build the same template again, how often does the recipe change — always custom, sometimes, rarely?

Q3. When the recipe changes for one specific order (e.g., upgraded display for Aarhus), is that: (a) a change for THIS order only, (b) a new permanent version of the template, or (c) both happen?

Q4. Do you have a naming scheme in mind for templates? ("Hospital Service Bike v1", "HSB-2026-A", or your own pattern.)

Manufacturing orders

An MO is the build directive — what gets built, how many, by when. It's the thing your mechanics open at the start of the day.

QUESTIONS FOR DENNIS — Manufacturing orders

Q5. Do you ever build bikes WITHOUT a customer order — for stock, demo, or a trade show? How often does that happen?

Q6. Do you ever split one customer order into multiple MOs? (e.g., "20 bikes for delivery in June, 30 in August".)

Q7. Once an MO is started, do you ever change the target quantity mid-build? (Customer asks for 5 more; you scrap 3 defective frames.)

Q8. Do you assign each MO to a specific lead mechanic, or is it more "whoever is around"?

The actual build

These are the small but operationally important details — the ones that decide whether the system feels natural in the workshop or annoying.

QUESTIONS FOR DENNIS — Build mechanics

Q9. When does inventory get "used up" in your mind — when the bike build starts (50 sets of parts pulled at once), when each part gets installed, or when the whole bike is finally complete?

Q10. For e-bikes: are all 5 identifiers (frame, lock, battery lock, battery, charger) typically known when the bike is finished, or do some get added later (e.g., AirTag fitted by the customer themselves)?

Q11. Frame numbers — is there a pattern you'd like the system to enforce? ("JP-2026-HSB-001" style, or freeform whatever the supplier prints?)

Q12. Substitutions: when you swap part A for part B during a build, is it usually for ALL bikes in that order, or only one or two specific bikes within it?

Returns and restarts

Reality is messy. Frames turn out to have hairlines, mechanics realise mid-build the wrong cable went on, customers return bikes for refurbishment. We need the system to handle the unhappy paths without making you fight it.

QUESTIONS FOR DENNIS — Returns and restarts

Q13. If a bike is half-built and the frame turns out defective, how do you handle it today — scrap and start over, or strip the parts back and re-use them on a fresh frame?

Q14. Have you ever fully rebuilt a returned bike — same frame number, but completely re-assembled? Should that be tracked as a new build or as maintenance?

What we should call things

Names matter. The system uses generic terms today; if you have your own words for these, the system should use yours.

QUESTIONS FOR DENNIS — Vocabulary

Q15. "Manufacturing Order" — works for you, or do you prefer "Build Order", "Production Run", or your own term?

Q16. "Bike Template" — works, or do you prefer "Recipe", "Specification", "Build Sheet"? What do you actually say in conversation when you mean this?

What happens after you sign off

Once we have your answers, we will:

1. Lock in the design decisions, the same way Phase 1 was finalised.
2. Build Phase 2 in the same incremental, browser-verified way Phase 1 was built — small slices that each ship usable functionality, browser-tested before each one is called done.
3. Show you each piece as it lands, not in a giant reveal at the end.

The order of work in Phase 2 will be: **(a)** bike models & templates so you have somewhere to define recipes, then **(b)** the bikes themselves and their identifiers, then **(c)** the manufacturing-order workbench that ties it all together. Each part will be browser-verified end-to-end against the real database before the next part starts.

Estimated time for the full Phase 2: roughly 6–8 weeks for one developer working with AI tools, faster with a second pair of hands. Phase 3 (Customer/Fleet management) follows once Phase 2 is in real use.

Thank you for taking the time to review this. The whole system depends on us getting your business right.